Ivan Evdokimov, Ph.D.

Belfast, UK | ivevdm@gmail.com | linkedin.com/in/ivan-evdokimov | github.com/ivan020

Professional Summary

Python Developer and Software Engineer with 6+ years of experience delivering production-grade AI systems, LLM/RAG pipelines, and FastAPI microservices. Strong software fundamentals in Python, APIs, microservices, testing, and Git, with hands-on cloud infrastructure and CI/CD experience. Proven ability to collaborate with stakeholders, demonstrate clear communication and client readiness, and maintain a growth mindset in fast-paced environments.

Experience

Software Engineer – GenAI, University of Essex — Colchester, UK — Nov 2025 — Present

- Designed and built enterprise-scale FastAPI microservices and REST APIs to power AI-driven development workflows by implementing CI/CD pipelines with GitHub Actions and Vercel, enabling weekly release cycles.
- Developed automated testing frameworks to validate LLM-generated code by implementing comprehensive validation checks, reducing manual review time by 70% while maintaining production stability and security.
- Delivered end-to-end MLOps infrastructure for a multi-tenant SaaS platform by integrating monitoring, deployment, and retraining pipelines, accelerating feature delivery from monthly to weekly releases.

Software Engineer – GenAI, UK Data Service — Colchester, UK — Jan 2023 — Oct 2025

- Built and deployed FastAPI-based LLM and ML services for automated classification tasks using PyTorch, Hugging Face, vLLM, and Ollama, achieving 95%+ accuracy on sensitive datasets.
- Implemented RAG pipelines to improve metadata classification by integrating embeddings, prompts, and retrieval strategies, enhancing retrieval accuracy for large-scale datasets.
- Designed serverless ML infrastructure on AWS cloud platforms by automating retraining and monitoring pipelines with Lambda, Step Functions, and EFS, reducing manual workload by 50%.
- Collaborated with onshore and non-technical stakeholders to gather requirements by translating business needs into scalable technical solutions, demonstrating clear communication and client readiness in enterprise environments.

Software Engineer & ML Instructor, University of Essex — Colchester, UK — Oct 2021 — Dec 2022

- Migrated quantitative macroeconomic models from MATLAB to C++ by optimizing algorithms and data structures, reducing simulation runtime by 8x for cross-university research.
- Taught Data Structures, Machine Learning, and software engineering to 100+ students by designing interactive lectures and labs, developing strong written and verbal communication skills.

Education

University of Essex, PhD in Computational Finance — Oct 2021 — May 2025

- Built high-performance data pipelines in C++ for large-scale financial datasets from Bloomberg by implementing efficient data processing and cleaning workflows, enabling AI-driven model training.
- Engineered a PyTorch-based transfer learning framework for financial applications by adapting pre-trained models to financial data, enabling accurate financial time-series prediction and portfolio optimization.
- Applied ML and Bayesian approaches to develop trading strategies by analyzing fundamental financial data, providing data-driven insights for portfolio management.

University of Essex, MSc in Financial Econometrics — Oct 2020 — Sep 2021

- Built a C++ agent-based financial simulation modelling macroeconomic dynamics under negative interest rates, demonstrating low-latency financial systems experience.

Projects

FinBot: Chat-Based Trading Simulation — github.com/ivan020/FinBot

- Built a real-time trading simulation with microservices architecture (Go, PostgreSQL/Supabase) and API integrations that serves up to 120 users monthly.
- Developed a JavaScript UI for real-time portfolio performance visualisation with interactive graphs and tables.

Chat-LLM — github.com/ivan020/ChatLLM

- Built a self-hosted Ollama MCP server on Raspberry Pi 5 by implementing custom tools for AI-driven story generation, enabling automated content creation for Twitch viewers.
- Created integration for the streaming website to show images generated by stream viewers.

MTG Card Reader — https://mtg-front-end-nine.vercel.app/

- Built a Golang REST API with Mistral LLM integration by implementing automatic card recognition from photos, enabling real-time price lookup for 50+ monthly active users.
- Developed a React frontend for card cataloguing, filtering, and deck management — reaching 50 monthly active users.